








Blockchain Applications

Lecture 3: Volatility and Tail Risk



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
Master Blockchain for Business (MBB), FABIZ
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Today's Question and Route

- **Question**: how much can a crypto position lose tomorrow, and how do we know the risk model is right?
  - an exchange sets margins, a fund reports risk, a firm with crypto on its balance sheet needs a loss budget
- **Route**
  - from volatility clustering (Lecture 2) to volatility models: exponentially weighted moving average (EWMA), generalised autoregressive conditional heteroskedasticity (GARCH)
  - estimation by maximum likelihood with Student-t errors; diagnostics
  - Value-at-Risk (VaR) and Expected Shortfall (ES): four methods
  - backtesting: Kupiec, Christoffersen, the Basel traffic light, an ES check
  - out of sample: the forecasts meet new data
- Data: daily log returns in %, 1 January 2018 – 18 September 2026 (EODHD)

Learning Outcomes

- Write down EWMA and GARCH(1,1) and interpret ω, α, β, persistence and half-life
- Read a GARCH estimation table with Student-t errors and judge the model with residual diagnostics
- Define VaR and ES, and compare Normal, historical, EWMA and GARCH-t estimates
- Interpret the Kupiec, Christoffersen and conditional-coverage tests and the Basel traffic light
- Judge what a short out-of-sample period can and cannot show

Abbreviations

AIC Akaike information criterion
ATS Advanced Time Series Analysis and Forecasting (course)
BTC Bitcoin
CDF cumulative distribution function
DM Diebold–Mariano (test)
EODHD EOD Historical Data (data vendor)
ES expected shortfall
ETF exchange-traded fund
EWMA exponentially weighted moving average
FABIZ Faculty of Business Administration in Foreign Languages, ASE
GARCH generalised autoregressive conditional heteroskedasticity
GARCH-t GARCH with Student-t errors
GJR Glosten–Jagannathan–Runkle (asymmetric GARCH)
HAC heteroskedasticity- and autocorrelation-consistent
HS historical simulation
IGARCH integrated GARCH
i.i.d. independent and identically distributed
KS Kolmogorov–Smirnov (test)
LR likelihood ratio
MAR market risk standard of the Basel Framework
MSc Master of Science
MV minimum variance
PIT probability integral transform
QLIKE quasi-likelihood loss
RC risk contribution
RMS root mean square
S&P 500 Standard & Poor's 500 stock index
sd standard deviation
se standard error
TLT iShares 20+ Year Treasury Bond ETF
US United States
USD US dollar
VaR value-at-risk

Reading and Tools

- Volatility models: Engle (1982), Bollerslev (1986), Glosten, Jagannathan & Runkle (1993); for crypto: Katsiampa (2017), Chu et al. (2017)
- Risk measures and backtests: Artzner et al. (1999), Kupiec (1995), Christoffersen (1998), Basel Committee, MAR33
- Video unit on Quantinar: Robust Markowitz for Crypto-Assets
- Related lectures: Advanced Time Series Analysis and Forecasting (ATS), *GARCH Models* and *VaR & ES* (2025); MSc dissertation A.-T. Ginavar, *Risk models for digital assets* (VaR/ES backtests for six cryptos)
- Estimation: Python, package `arch`; all numbers on the slides are re-estimated for this lecture

From Clustering to a Model

- Lecture 2: returns are hardly autocorrelated, squared returns are strongly autocorrelated
- Model the return as a mean plus a shock with time-varying scale:

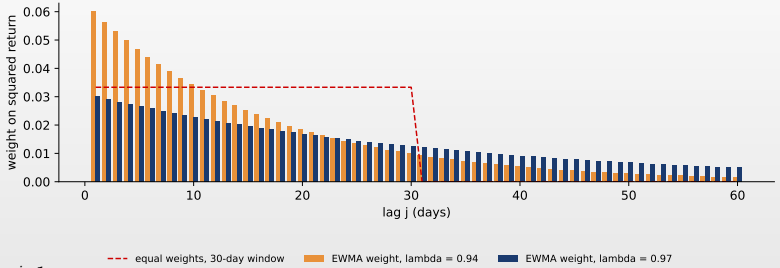
$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad z_t \overset{\text{i.i.d.}}{\sim} (0,1)$$

  - $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$: the **conditional variance**, known at the end of day $t-1$
  - $\mathcal{F}_{t-1}$: the information up to day $t-1$
- **EWMA** (RiskMetrics): an exponentially weighted average of past squared shocks

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,\varepsilon_{t-1}^2 = (1-\lambda)\sum_{j\geq 1}\lambda^{j-1}\varepsilon_{t-j}^2, \qquad \lambda = 0.94$$

  - weights decay geometrically; effective memory $1/(1-\lambda) \approx 17$ days
  - no long-run level: after a shock, volatility never returns to a mean

How EWMA Weights the Past



- Weight $(1-\lambda)\lambda^{j-1}$ on the squared return j days ago; red: a 30-day equal-weight window
- $\lambda = 0.94$: 6% on yesterday, 1.0% on day 30 (0.06×0.94^{29}); the first 30 days carry $1 - 0.94^{30} = 84\%$ of the weight
- No cliff at the end of the window: a crash 31 days ago does not suddenly drop out, as it does with a rolling standard deviation

GARCH(1,1)

- Bollerslev (1986):

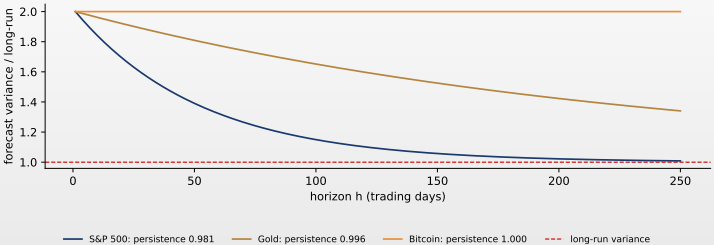
$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0,\ \alpha, \beta \geq 0$$

  - α: reaction to yesterday's news; β: memory of yesterday's variance; $\alpha + \beta$: **persistence**
- If $\alpha + \beta < 1$: covariance-stationary, with long-run (unconditional) variance

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}, \qquad \mathrm{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}\left(\sigma_{t+1}^2 - \bar{\sigma}^2\right)$$

- **Half-life** of a volatility shock: $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$ days
- $\alpha + \beta = 1$: **integrated** GARCH (IGARCH); shocks never die out; EWMA is the case $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$

After a Shock: Volatility Forecasts by Horizon



- Forecast variance $\mathrm{E}_t[\sigma^2_{t+h}]$ relative to the long-run level, after a shock that doubles today's variance, with the estimated parameters
- S&P 500 ($\hat{\alpha} + \hat{\beta} = 0.981$): half-life $\ln 0.5 / \ln 0.981 = 36.1$ days; after 250 days the shock is gone
- Gold (0.996): still 34% above the long run after a year; Bitcoin (1.000): the shock never decays, $\sqrt{h}$ scaling of today's volatility

Student-t Errors and Maximum Likelihood

- Heavy tails remain after GARCH filtering: let z_t follow a Student-t with $\nu > 2$ degrees of freedom, scaled to variance 1
- Log-likelihood of $\theta = (\mu, \omega, \alpha, \beta, \nu)$:

$$\ell(\theta) = \sum_{t=1}^{n} \left[\ln \Gamma\left(\tfrac{\nu+1}{2}\right) - \ln \Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2} \ln\left(\pi(\nu-2)\sigma_t^2\right) - \tfrac{\nu+1}{2} \ln\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right) \right]$$

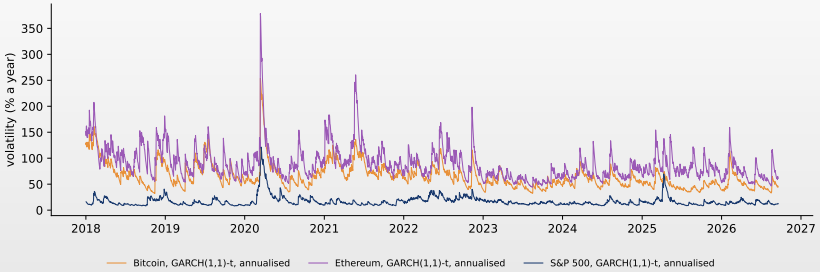
  - σ_t^2 computed recursively from θ; $\hat{\theta} = \arg\max \ell(\theta)$ by numerical optimisation
  - standard errors: robust (sandwich) covariance of $\hat{\theta}$
  - $\nu \to \infty$: the Normal; ν close to 2: very heavy tails

GARCH(1,1)-t Estimates, 2018–2026

	n	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	half-life (days)	$\hat{\nu}$
Bitcoin	3 183	0.144 (0.058)	0.076 (0.012)	0.924 (0.012)	1.000	∞	3.13 (0.15)
Ethereum	3 183	0.550 (0.373)	0.118 (0.038)	0.883 (0.042)	1.000	∞	3.19 (0.20)
S&P 500	2 190	0.035 (0.009)	0.162 (0.022)	0.819 (0.021)	0.981	36	5.85 (0.71)
Gold	2 634	0.013 (0.006)	0.053 (0.014)	0.942 (0.014)	0.996	160	3.33 (0.25)

- Daily log returns in %; constant mean; robust standard errors in brackets
- Interpretation
  - crypto: $\hat{\alpha}+\hat{\beta}$ at the boundary 1, an IGARCH; no finite long-run variance, no mean reversion of volatility in this sample
  - S&P 500: a shock to volatility halves in 36 trading days; long-run volatility $\sqrt{252\,\hat{\omega}/(1-\hat{\alpha}-\hat{\beta})} = 21.6\%$
  - $\hat{\nu} \approx 3.1$ for crypto, 5.9 for the S&P 500: crypto shocks are heavy-tailed even after filtering

Conditional Volatility



- $\hat{\sigma}_t\sqrt{A}$ from GARCH(1,1)-t, $A = 365$ for crypto, 252 for the S&P 500
- Peaks: Bitcoin 253.5% (13 March 2020), Ethereum 378.4%, S&P 500 121.4% (17 March 2020)
- 18 September 2026: Bitcoin 44.6%, Ethereum 61.2%, S&P 500 12.7%; Bitcoin median 59.2%

Diagnostics: Did the Model Remove the Clustering?

- Standardised residuals $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$ should be i.i.d.: test $\hat{z}_t^2$ with Ljung–Box $Q(10) \sim \chi^2_{10}$ (5% critical value 18.31)

	$Q(10)$ of r_t^2	$Q(10)$ of $\hat{z}_t^2$	p-value for $\hat{z}_t^2$
Bitcoin	97.6	5.5	0.855
Ethereum	131.8	8.8	0.552
S&P 500	2 338.8	12.0	0.283
Gold	440.9	22.9	0.011

- Interpretation
  - for crypto and the S&P 500, GARCH removes the volatility clustering: no dependence left in $\hat{z}_t^2$
  - gold: some dependence remains (p 0.011); a richer model (asymmetry, regime changes in 2025–2026) is needed
  - sample 2018–2026; the $Q(10) = 212.7$ of Lecture 2 used the longer sample since 2014

VaR and ES: Definitions

- For a one-day return r_{t+1} with conditional distribution F_t and level p (e.g. 1%):

$$\mathrm{VaR}_p = -F_t^{-1}(p), \qquad \mathrm{ES}_p = -\mathrm{E}\big[r_{t+1} \mid r_{t+1} \le -\mathrm{VaR}_p\big] = \frac{1}{p}\int_0^p \mathrm{VaR}_u \, du$$

  - VaR: the loss exceeded with probability p; ES: the average loss when VaR is exceeded
- Artzner et al. (1999): a **coherent** risk measure is monotone, translation-invariant, positively homogeneous and **subadditive**: $\rho(X+Y) \le \rho(X) + \rho(Y)$
  - ES is coherent; VaR is not subadditive in general, so diversification can appear to raise VaR
  - ES sees the size of the tail; VaR only its threshold
- Regulation (Basel Committee, MAR33): ES at 97.5% replaced VaR at 99% for market risk
  - under the Normal: $\mathrm{VaR}_{1\%} = 2.326\,\sigma$ and $\mathrm{ES}_{2.5\%} = 2.338\,\sigma$, almost equal; under heavy tails ES is larger

Four Ways to Compute VaR and ES

- **Normal**: $\mathrm{VaR}_p = -(\hat{\mu} + \hat{\sigma}\,\Phi^{-1}(p))$, $\mathrm{ES}_p = -\hat{\mu} + \hat{\sigma}\,\varphi(\Phi^{-1}(p))/p$
- **Historical simulation** (HS): $\mathrm{VaR}_p = -\hat{q}_p$, the empirical p-quantile of the last W returns; ES_p = minus the mean of returns below $\hat{q}_p$
- **EWMA**: Normal formulas with $\hat{\sigma}_{t+1}$ from the EWMA recursion, $\hat{\mu} = 0$
- **GARCH-t**: with $q = t_\nu^{-1}(p)$ and $s = \sqrt{(\nu-2)/\nu}$

$$\mathrm{VaR}_p = -\big(\hat{\mu} + \hat{\sigma}_{t+1}\, s\, q\big), \qquad \mathrm{ES}_p = -\hat{\mu} + \hat{\sigma}_{t+1}\, s\, \frac{f_\nu(q)}{p}\,\frac{\nu + q^2}{\nu - 1}$$

  - f_ν: the Student-t density; s rescales the t to variance 1
- **Horizon**: $\mathrm{VaR}_h \approx \sqrt{h}\,\mathrm{VaR}_1$ only for i.i.d. Normal returns; with GARCH, use the variance forecasts $\mathrm{E}_t[\sigma^2_{t+j}]$

Worked Example: GARCH-t VaR and ES for Bitcoin

- Inputs on 18 September 2026: $\hat{\sigma}_{t+1} = 2.761\%$, $\hat{\mu} = 0.06\%$, $\hat{\nu} = 3.126$
- Scale factor $s = \sqrt{(\nu-2)/\nu} = \sqrt{1.126/3.126} = 0.600$
- VaR 1%: $q = t_{3.126}^{-1}(0.01) = -4.398$

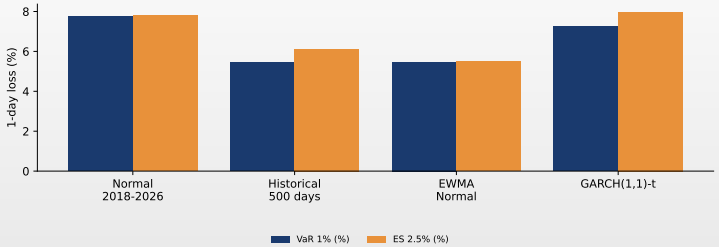
$$\mathrm{VaR}_{1\%} = -0.06 + 2.761 \times 0.600 \times 4.398 = 7.23\%$$

- ES 2.5%: $q = t_{3.126}^{-1}(0.025) = -3.111$; $\dfrac{f_\nu(q)}{p}\dfrac{\nu+q^2}{\nu-1} = 4.844$

$$\mathrm{ES}_{2.5\%} = -0.06 + 2.761 \times 0.600 \times 4.844 = 7.97\%$$

- Under the Normal with the same σ: $2.761 \times 2.338 = 6.46\%$; the t tail adds 1.5 percentage points

VaR and ES by Method



- Bitcoin, one-day horizon, forecasts for 19 September 2026 (table on the next slide)
- Normal and EWMA: ES $\approx$ VaR; historical and GARCH-t: ES exceeds VaR by 13% and 10%

Bitcoin: VaR and ES for 19 September 2026

Method	VaR 1% (%)	ES 2.5% (%)	VaR, USD per 1m	ES, USD per 1m
Normal, 2018–2026	7.75	7.79	74 567	74 923
Historical simulation, 500 days	5.43	6.12	52 854	59 360
EWMA, $\lambda = 0.94$, Normal	5.47	5.49	53 208	53 463
GARCH(1,1)-t	7.23	7.97	69 753	76 614

- One-day losses of a long position, in log-return % and in USD for a position of 1 million USD ($10^{6}(1 - e^{-x/100})$)
- GARCH forecast: $\hat{\sigma}_{t+1} = 2.76\%$ a day
- Interpretation
  - the Normal and EWMA give $\mathrm{ES}_{2.5\%} \approx \mathrm{VaR}_{1\%}$: they ignore the heavy tail
  - GARCH-t: ES exceeds VaR by 10%, the cost of the heavy tail ($\hat{\nu} = 3.13$)
  - the four VaR figures differ by up to 43%: the choice of model is itself a risk (**model risk**)

Backtesting VaR: the Hit Sequence

- Hit (violation): $I_t = \mathbb{1}\{r_t < -\mathrm{VaR}_{p,t}\}$; a correct model has $I_t \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$
- **Unconditional coverage** Kupiec (1995): x hits in n days, $\hat{\pi} = x/n$

$$\mathrm{LR}_{uc} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x} \sim \chi^2_1$$

- **Independence** Christoffersen (1998): n_{ij} = number of days in state j after state i; $\hat{\pi}_{ij} = n_{ij}/(n_{i0} + n_{i1})$

$$\mathrm{LR}_{ind} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1-\hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \sim \chi^2_1$$

- **Conditional coverage**: $\mathrm{LR}_{cc} = \mathrm{LR}_{uc} + \mathrm{LR}_{ind} \sim \chi^2_2$
- **Basel traffic light**, 250 days at 1%: green 0–4 hits, yellow 5–9, red 10 or more

Worked Example: the Kupiec Test

- Normal VaR on a 250-day window, Bitcoin 2020–2026: $n = 2\,453$ days, $x = 45$ hits, $p = 0.01$, $\hat{\pi} = 45/2\,453 = 0.0183$

$$\ln L_0 = 2\,408 \ln 0.99 + 45 \ln 0.01 = -231.4, \qquad \ln L_1 = 2\,408 \ln(1 - 0.0183) + 45 \ln 0.0183 = -224.5$$

$$\mathrm{LR}_{uc} = -2(\ln L_0 - \ln L_1) = 13.84 > 3.84$$

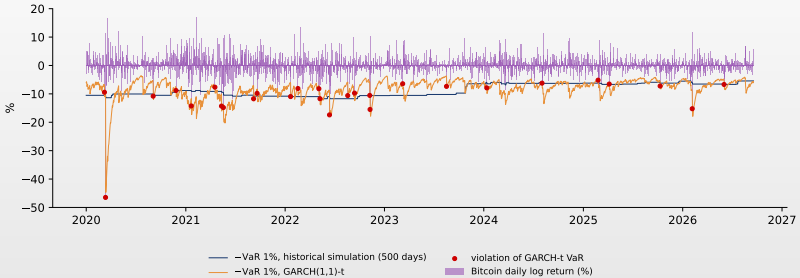
- The model is rejected: 45 hits where 24.5 were expected, almost twice the promised frequency
- For a bank this is a capital issue: Basel multiplies market-risk capital by a factor that rises with the number of exceptions

Backtest: Bitcoin, 1-Day VaR 1%, 2020–2026

Method	hits	rate (%)	LR_{uc} (p)	LR_{ind} (p)	LR_{cc} (p)	last 250	zone	mean VaR (%)
Normal, 250 days	45	1.83	13.84 (<0.001)	0.04 (0.850)	13.88 (0.001)	5	yellow	7.16
HS, 500 days	25	1.02	0.01 (0.924)	1.29 (0.256)	1.30 (0.523)	3	green	8.64
EWMA, Normal	51	2.08	22.01 (<0.001)	0.00 (0.952)	22.01 (<0.001)	5	yellow	6.78
GARCH(1,1)-t	28	1.14	0.47 (0.491)	0.95 (0.329)	1.43 (0.490)	2	green	8.12

- 2 453 out-of-sample days, 1 January 2020 – 18 September 2026; expected hits $np = 24.5$
- GARCH-t re-estimated every 21 days on all data up to the previous day, then filtered daily
- Interpretation
  - Normal and EWMA fail unconditional coverage: almost twice as many hits as promised
  - HS and GARCH-t pass all three tests; GARCH-t reacts to news, HS needs a higher average VaR (8.64% vs 8.12%) for the same coverage

Violations Through Time



- The HS line moves in steps and lags the market; the GARCH-t line widens within days after a shock
- Hits of GARCH-t are spread over time: no clustering, in line with LR_{ind} (p 0.329)

Checking Expected Shortfall

- ES is not elicitable alone; a simple check in the spirit of McNeil–Frey: on the days beyond the 2.5% quantile, compare realised loss and predicted ES

$$\bar{R} = \frac{1}{N} \sum_{t:\, r_t < -\mathrm{VaR}_{2.5\%,t}} \frac{-r_t}{\mathrm{ES}_{2.5\%,t}}, \qquad \text{correct model: } \bar{R} \approx 1$$

Method	tail days	rate (%)	mean loss (%)	mean ES (%)	$\bar{R}$
Normal, 250 days	71	2.89	8.48	6.72	1.26
HS, 500 days	61	2.49	9.05	9.14	1.00
EWMA, Normal	76	3.10	7.94	6.11	1.31
GARCH(1,1)-t	76	3.10	7.92	8.04	0.97

- Normal-based ES: realised tail losses are 26–31% larger than predicted; HS and GARCH-t are close to 1
- GARCH-t has too many days beyond its 2.5% quantile (3.10%), but the loss size on those days is right

The Forecasts Meet New Data

- All estimates in this lecture use data to 18 September 2026; EODHD data to 23 October 2026 are now available
- Design: parameters **frozen** on 18 September, nothing re-estimated; only the filter is updated with new returns

$$\hat{\sigma}^2_{t|t-1} = \hat{\omega} + \hat{\alpha}\,(r_{t-1} - \hat{\mu})^2 + \hat{\beta}\,\hat{\sigma}^2_{t-1|t-2}, \qquad t = T+1, \ldots, T+H$$

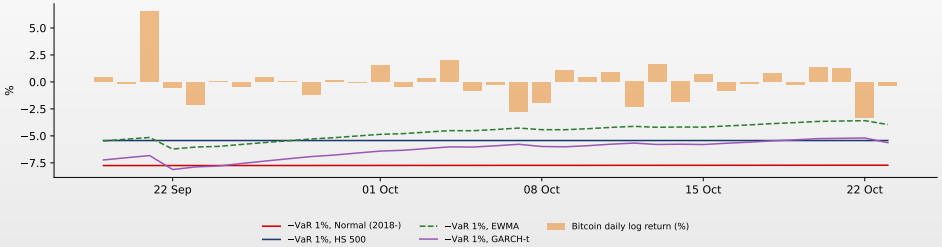
- Two questions
  - Was the *level* of volatility right? realised $\sqrt{A\,\overline{r^2}}$ against the 18 September forecast $\sqrt{A\,\frac{1}{H}\sum_h E_T \sigma^2_{T+h}}$
  - Did the VaR hold? hits against *np*
- 35 calendar days for crypto and gold, 25 trading days for the S&P 500: a short window, so the question is also *how much can 35 days tell us?*

Volatility: Forecast against Realised

annualised, %	Bitcoin	Ethereum	S&P 500	Gold
return over the window (%)	0.2	−9.1	−0.4	−9.3
GARCH-t $\hat{\sigma}_{T+\mathbf{1}}$, 18 Sep	52.7	72.6	11.9	20.5
GARCH-t, average over the window	60.6	93.2	14.4	21.1
EWMA, 18 Sep	44.9	58.8	10.4	19.7
Normal, 2018–2026	64.1	84.3	19.2	15.7
realised, 19 Sep – 23 Oct	31.8	36.1	10.3	15.7
realised / GARCH forecast	0.52	0.39	0.72	0.74

- Every model over-predicted: Bitcoin's realised 31.8% equals the calm of August 2026, while the GARCH forecast reverts to a long-run level near 60%
- Bitcoin and Ethereum have $\hat{\alpha} + \hat{\beta} \geq 1$: the 18 September forecast path *rises* towards the long-run level instead of decaying
- Gold fell 9.3% with ordinary volatility (15.7%): a large price move is not the same as high volatility

Bitcoin VaR, Rolled Forward



1-day VaR 1%, 35 days	Normal	HS 500	EWMA	GARCH-t
mean VaR (%)	7.73	5.43	4.66	6.28
mean capital per 1m USD	74 428	52 854	45 500	60 871
hits (expected 0.35)	0	0	0	0

- Worst day −3.27% (22 October): 0.63 of that day's GARCH-t VaR; 18 September forecast for 19 September: VaR 7.23%, realised +0.41%

How Much Can 35 Days Tell Us?

- With a correct 1% VaR, the number of hits is $X \sim \mathrm{Bin}(35, 0.01)$:

$$P(X = 0) = 0.99^{35} = 0.703, \qquad \mathrm{LR}_{uc}(X = 0) = -2 \cdot 35 \ln 0.99 = 0.70 \; (p = 0.40)$$

- A model with a *true* hit rate of 2% (as Normal-250 in the backtest) still shows zero hits with probability $0.98^{35} = 0.493$
- Power: to detect 2% against 1% with probability 0.8 at the 5% level one needs about **910 days**
- Conclusions
  - Zero hits in five weeks is *no evidence* for any of the four methods: all four "pass"
  - The methods differ in capital: GARCH-t asks for 34% more than EWMA in this window; the long backtest, not this window, says which one is right
  - Use the whole density, not only the hits (Appendix)

Summary

- GARCH(1,1)-t fits crypto with $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH) and $\hat{\nu} \approx 3.1$: shocks are very heavy-tailed even after filtering
- The model removes the clustering: $Q(10)$ of $\hat{z}_t^2 = 5.5$ (p 0.855) for Bitcoin
- Bitcoin 1-day VaR 1% on 19 September 2026: 5.4% to 7.8% depending on the method; ES 2.5% up to 8.0%
- Backtest 2020–2026: Normal and EWMA fail Kupiec (45 and 51 hits for 24.5 expected); HS and GARCH-t pass all tests
- With Normal-based ES, realised tail losses are 26–31% larger than predicted
- Out of sample (to 23 October 2026) all models over-predicted volatility; 35 days cannot discriminate between VaR methods

Interpretation Exercises

1. Bitcoin: $\hat{\alpha} + \hat{\beta} = 1.000$; S&P 500: 0.981. What does each imply for a 10-day volatility forecast made after a crash?
2. Ljung–Box for gold: $Q(10)$ of $\hat{z}_t^2 = 22.9$ (p 0.011). Is the model adequate? What would you try next?
3. A risk manager uses EWMA-Normal for Bitcoin and had 5 violations in the last 250 days. Which zone is that, and what do the full-period tests say about the model?
4. The GARCH-t ES for 1 million USD is 76 614 USD and the Normal ES is 74 923 USD, yet the backtest favours GARCH-t clearly. Explain using $\bar{R}$.
5. Out of sample (19 Sep – 23 Oct 2026) no method had a Bitcoin VaR hit. A manager concludes that the Normal model is now fine. Respond with the binomial argument.

⊡ Next: Lecture 4, stablecoins and the measurement of depeg risk

Appendix: Further Material

Optional material for self-study. It is **not examined**.

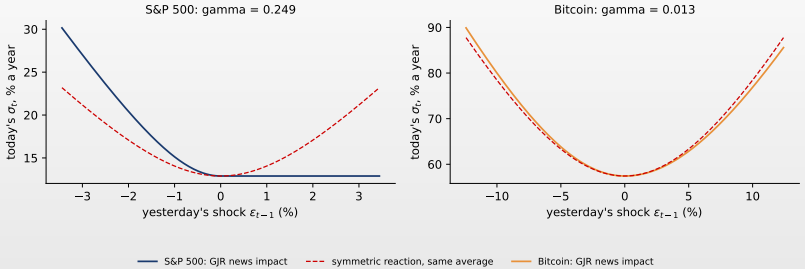
- Is There a Leverage Effect?
- News Impact Curves
- Diagnostics: Normal or Student-t Errors?
- The Traffic Light Through Time
- Euler Risk Contributions
- Worked Example: Euler Contributions with Two Assets
- A 60/40 Portfolio with Bitcoin
- Risk Share against Weight
- Does Bitcoin Improve the Efficient Frontier?
- One-Step Volatility Forecasts, Day by Day
- Which Forecast Is Best? QLIKE and Diebold–Mariano
- Was the Density Right? PIT and the Berkowitz Test
- The Portfolio: Correlation Is the Unstable Input

Is There a Leverage Effect?

	$\hat{\gamma}$ (GJR)	$t(\hat{\gamma})$	LR $= 2(\ell_{\mathbf{GJR}} - \ell_{\mathbf{GARCH}})$	p-value	AIC: GARCH / GJR
Bitcoin	0.014	0.72	0.68	0.408	15 518 / 15 520
Ethereum	0.028	0.74	1.19	0.275	17 493 / 17 494
S&P 500	0.249	6.25	75.89	< 0.001	5 902 / 5 828
Gold	-0.056	-3.22	14.76	< 0.001	6 555 / 6 543

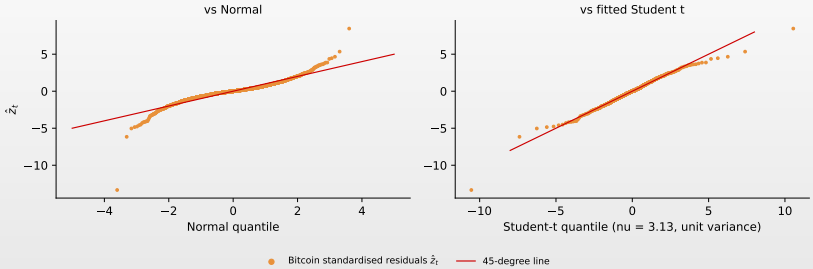
- 5% critical value of χ^2_1: 3.84
- Interpretation
  - S&P 500: falls raise volatility much more than rises; GJR preferred by LR and the Akaike information criterion (AIC)
  - gold: $\hat{\gamma} < 0$: rises raise volatility more than falls, the pattern of a safe-haven asset
  - crypto: no asymmetry; GARCH(1,1) is enough, AIC prefers it

News Impact Curves



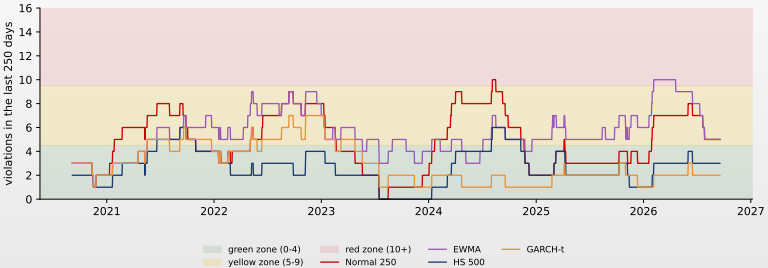
- Today's volatility as a function of yesterday's shock, with yesterday's variance at its sample median (GJR-GARCH-t)
- S&P 500: a -3% day raises volatility to about 28% a year, a $+3\%$ day leaves it at 13%: strong asymmetry
- Bitcoin: the curve is almost symmetric ($\hat{\gamma} = 0.013$): good and bad news raise volatility alike

Diagnostics: Normal or Student-t Errors?



- Bitcoin $\hat{z}_t$ against Normal quantiles (left) and against the fitted Student-t with $\hat{\nu} = 3.13$ (right)
- The Normal fails in both tails; the Student-t fits well except the single most extreme day (12 March 2020: $\hat{z}_t = -13.4$ against a t quantile of -10.5)

The Traffic Light Through Time



- Number of hits in the last 250 days for each method; background: Basel green, yellow and red zones
- Share of days in the yellow or red zone: Normal 50.3%, EWMA 75.6% (red 3.4%), historical 10.5%, GARCH-t 22.9% (never red)
- A model can pass a full-sample test and still spend long periods in the yellow zone: monitor it continuously

Euler Risk Contributions

- Portfolio return $r_p = w^\top r$, weights w, covariance matrix Σ; volatility $\sigma_p = \sqrt{w^\top \Sigma w}$
- σ_p is homogeneous of degree 1 in w, so by Euler's theorem it is the sum of the risk contributions (RC)

$$\sigma_p = \sum_i w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_i \mathrm{RC}_i, \qquad \mathrm{RC}_i = \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- The same holds for ES; with historical data

$$\mathrm{ES}_p = \sum_i \mathrm{RC}_i^{ES}, \qquad \mathrm{RC}_i^{ES} = -\mathrm{E}\big[w_i r_i \mid r_p \le q_p(r_p)\big]$$

- RC_i/σ_p: the share of risk due to asset i; compare it with the weight w_i
- Minimum-variance (MV) portfolio: $w^{\mathrm{MV}} = \Sigma^{-1}1/(1^\top \Sigma^{-1}1)$

Worked Example: Euler Contributions with Two Assets

- 95% S&P 500 ($\sigma_1 = 19.2\%$), 5% Bitcoin ($\sigma_2 = 64.0\%$), correlation $\rho = 0.30$

$$\sigma_p^2 = 0.95^2 \cdot 19.2^2 + 0.05^2 \cdot 64.0^2 + 2 \cdot 0.95 \cdot 0.05 \cdot 0.30 \cdot 19.2 \cdot 64.0 = 377.8, \quad \sigma_p = 19.44\%$$

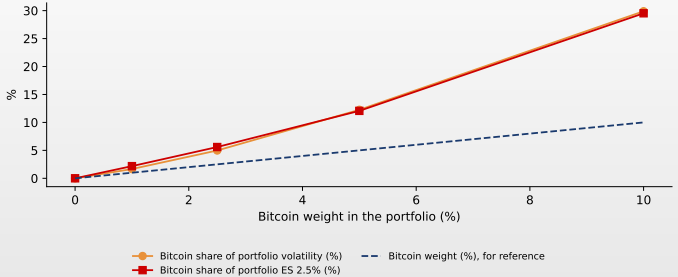
- Contributions $\mathrm{RC}_i = w_i(\Sigma w)_i / \sigma_p$
  - S&P 500: 18.01 percentage points (92.7%); Bitcoin: 1.43 points (7.3%)
  - check: $18.01 + 1.43 = 19.44 = \sigma_p$
- Bitcoin carries 7.3% of the risk with 5% of the money; in the 60/40 portfolio (next slide) it carries 12%, because Treasuries diversify the equity risk but not the Bitcoin risk

A 60/40 Portfolio with Bitcoin

Bitcoin weight (%)	volatility (% a year)	ES 2.5% (%, daily)	Bitcoin share of volatility (%)	Bitcoin share of ES (%)
0	12.28	2.32	0.0	0.0
1	12.35	2.35	1.7	2.2
2.5	12.51	2.39	5.0	5.6
5	12.91	2.48	12.3	12.1
10	14.20	2.70	29.9	29.5

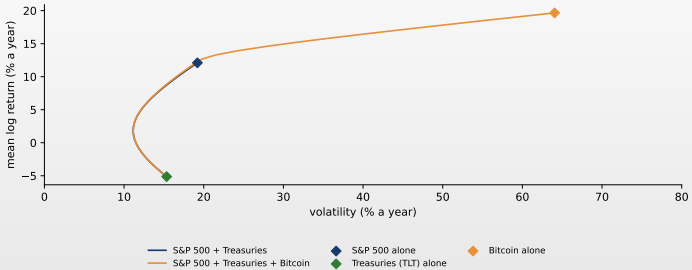
- S&P 500 and long US Treasuries (exchange-traded fund (ETF) TLT) in proportion 60/40, plus Bitcoin; 2 190 common days, 2018–2026
- Correlations: Bitcoin–S&P 500 0.30, Bitcoin–Treasuries −0.02, S&P 500–Treasuries −0.14
- Interpretation
  - a 5% Bitcoin weight carries 12% of the portfolio risk; 10% carries 30%
  - minimum-variance weights: S&P 500 40.6%, Treasuries 59.7%, Bitcoin −0.3%: Bitcoin does not reduce risk

Risk Share against Weight



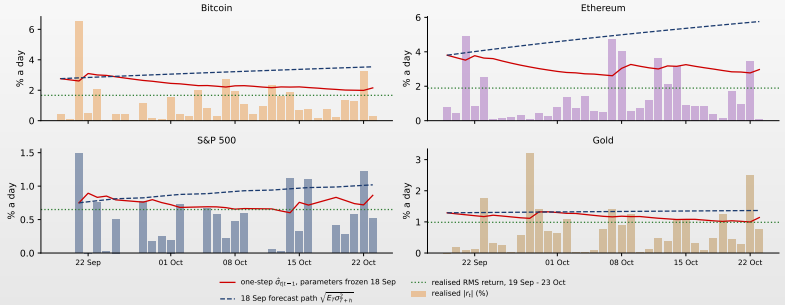
- The risk share grows faster than the weight: from 1.7 times the weight at 1% to 3 times at 10%
- A position limit for crypto should be set on its risk contribution, not on its weight
- Robust alternatives to the sample covariance: Robust Markowitz for Crypto-Assets

Does Bitcoin Improve the Efficient Frontier?



- Long-only minimum-variance frontiers, 2018–2026 common days; annualised mean log return and volatility
- Bitcoin alone: 19.7% a year at 64.0% volatility; S&P 500: 12.1% at 19.2%; Treasuries: −5.1% at 15.3%
- Adding Bitcoin extends the frontier only at high volatility; at the low-risk end the two frontiers coincide: in-sample means, so an upper bound on the benefit

One-Step Volatility Forecasts, Day by Day



- Red: GARCH-t with frozen parameters reacts to each $|r_t|$ (Bitcoin +6.5% on 21 September); blue: the 18 September term structure
- Ethereum's path goes up linearly: with $\hat{\alpha} + \hat{\beta} = 1.001$ the forecast never mean-reverts

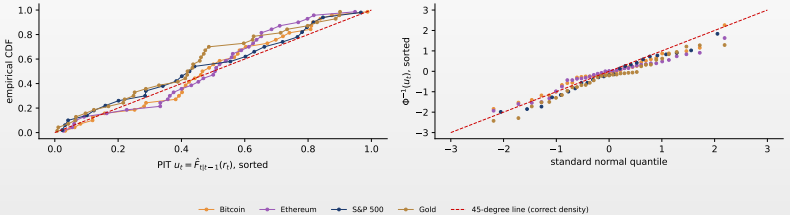
Which Forecast Is Best? QLIKE and Diebold–Mariano

- Loss robust to the noise in r_t^2 as a proxy of σ_t^2 (Patton, 2011): $L_t(h) = \ln h_t + r_t^2/h_t$
- $d_t = L_t(\text{GARCH}) - L_t(\text{other})$, $\mathrm{DM} = \bar{d}/\widehat{\mathrm{se}}_{\mathrm{HAC}}(\bar{d}) \sim N(0,1)$, HAC: heteroskedasticity- and autocorrelation-consistent; $\bar{d} > 0$: GARCH loses

	Bitcoin	Ethereum	S&P 500	Gold
mean QLIKE: GARCH-t	2.232	2.655	0.220	1.083
mean QLIKE: EWMA	2.094	2.435	0.184	1.088
mean QLIKE: constant (Normal)	2.667	3.152	0.670	0.972
GARCH vs EWMA: $\bar{d}$ (DM)	0.138 (1.97)	0.220 (2.99)	0.036 (0.83)	−0.005 (−0.10)
GARCH vs constant: $\bar{d}$ (DM)	−0.435 (−4.11)	−0.497 (−6.12)	−0.450 (−3.50)	0.110 (1.64)

- A time-varying forecast beats the constant one clearly for Bitcoin, Ethereum and the S&P 500
- EWMA beats GARCH for Ethereum ($\mathrm{DM} = 2.99$) and, at the margin, for Bitcoin: it forgets faster, and after a calm month that helps
- But in the 2020–2026 backtest EWMA-Normal failed the VaR test: being good at the *level* is not being good at the *tail*

Was the Density Right? PIT and the Berkowitz Test



- $u_t = \hat{F}_{t|t-\mathbf{1}}(r_t)$ is iid $U(0,1)$ if the density is right; $x_t = \Phi^{-\mathbf{1}}(u_t)$: fit $x_t = c + \rho x_{t-\mathbf{1}} + \sigma\varepsilon_t$, $\mathrm{LR} = 2(\ell_{\mathbf{1}} - \ell_{\mathbf{0}}) \sim \chi^{\mathbf{2}}_{\mathbf{3}}$ under $c = 0, \rho = 0, \sigma = 1$

	Bitcoin	Ethereum	S&P 500	Gold
sd of x_t (should be 1)	0.84	0.78	0.98	0.93
Berkowitz LR (p)	3.16 (0.368)	4.80 (0.187)	2.36 (0.501)	5.00 (0.172)
Kolmogorov–Smirnov D (p)	0.125 (0.604)	0.176 (0.203)	0.119 (0.833)	0.228 (0.045)

- sd of x_t below 1 for crypto: the predicted densities were too wide, in line with the over-predicted volatility; not significant in 35 days
- Gold: the Kolmogorov–Smirnov (KS) test rejects at 5%: mean of x_t is -0.31, the forecasts missed the downward drift, and one 1% hit (-3.18% on 28 September, $z = -2.9$)

The Portfolio: Correlation Is the Unstable Input

60/40 S&P 500/TLT plus Bitcoin Bitcoin weight	volatility (%, ann.) predicted	 realised	Bitcoin share of risk (%) predicted	 realised
0%	12.3	8.0	–	–
5%	12.9	8.7	12.3	13.0
10%	14.2	9.5	29.9	26.3

- Correlation Bitcoin – S&P 500: 0.30 in 2018–2026, 0.76 over the 25 new trading days
- Bitcoin's own volatility halved, yet its share of portfolio risk did not fall: the higher correlation offset the lower volatility

$$\text{share}_{\mathbf{BTC}} = \frac{w_B\,(\Sigma w)_B}{w'\Sigma w}, \qquad (\Sigma w)_B = w_B\sigma_B^2 + \textstyle\sum_{j\neq B} w_j \rho_{Bj}\sigma_B\sigma_j$$

- The risk budget is a forecast of the correlation, not only of the volatilities